Andrew Beasley

Formal Lab Report: Sample B

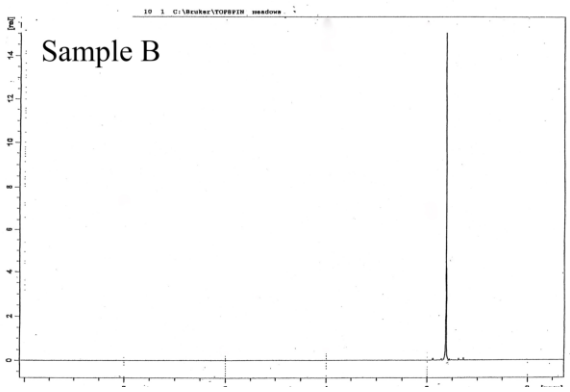
GTA: Dominic Fico

2081-001

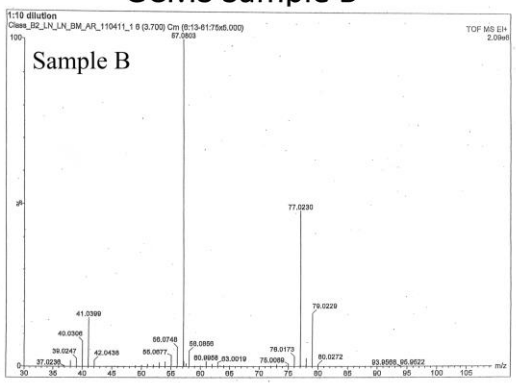
Gas Chromatography-Mass Spectrometry (GC-MS) separates compounds based on volatility, while Mass Spectrometry (MS) identifies components by analyzing mass-to-charge ratios. NMR exploits magnetic properties for structural insights, and FTIR measures infrared absorption to identify functional groups. GC-MS analyzes mixtures, NMR provides structural details, and FTIR identifies functional groups, offering a comprehensive understanding of a substance's composition and structure.

Observations:

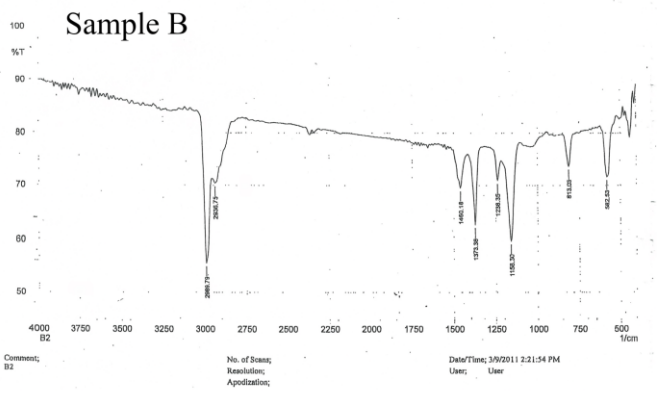
NMR Sample B



GCMS Sample B



FTIR Sample B



Conclusion:

The chemical analyzed by FTIR, NMR and GCMS was found to be tert-butylchloride. In the NMR spectrum, a lone peak emerges at around 1.6 ppm. In comparison, primary proton peaks manifest around 1.2-1.3 ppm, signaling a chemical downfield shift likely induced by a chloride atom. The singular peak denotes a singular proton type, with the heightened signal suggesting the involvement of multiple protons. In the FTIR analysis, the absence of activity above 3000 cm-1 rules out an alcohol group. Peaks at 2989.79 and 2936.75 confirm the presence of C-H bonds, and the lack of signals between 1600-1700 cm-1 eliminates the possibility of double or triple bonds. Signals between 1500-100 cm-1 identify a methyl group, while peaks at 813.03 and 582.83 corroborate the existence of a C-Cl bond. In GC-MS, the prominence of the base peak at 57.0803 m/z suggests the most stable cation with a chemical formula of C4H9. The (M+1)+ peak at 77.0230 and the (M+2)+ peak at 79.0229, with the latter roughly 35% the height of the former, indicate the presence of a chlorine atom. This aligns with the characteristic chlorine isotopic distribution. Furthermore, tert-butylchloride's molecular ion peak 93.9568 coincides with the compound's expected molecular weight, reinforcing its identification.

References: Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (Al): Auburn University.

Klein, David. Organic Chemistry. https://platform.virdocs.com/read/1892025/4/#/4/2[f03-sec-0001]/6,/1:0,/1:0.